# FOCUS CLUTCH MASTER CYLINDER REPLACEMENT

**AFFECTED VEHICLES:** **CERTAIN 2000 FOCUS EQUIPPED WITH MANUAL TRANSMISSIONS**

## OVERVIEW

This procedure provides the detail to replace the clutch master cylinder and the clutch pedal return spring. The clutch master cylinder is attached to the clutch pedal bracket. The fluid tube connections for the master cylinder are located at the dash panel and are accessible from under the hood. The brake fluid reservoir, which is constructed with separate internal chambers, supplies fluid to the clutch control system.

Two different manual transaxles are available in the Focus. The MTX75 is available with the 2.0L Zetec engine. The IB5 is available with the SPI engine. Clutch master cylinder replacement is the same for both applications, although there are different bleed procedures for each. Be sure to follow the correct bleed procedure based on the vehicle you are working on.

### REMOVAL

1. Install a memory saver and disconnect the battery negative cable.
2. Using a suction gun or similar tool, drain the brake fluid reservoir to the MIN mark.
3. Disconnect the air cleaner outlet tube at the throttle body, the MAFS electrical connector and the crankcase vent hose. Then remove the air cleaner assembly from the engine compartment.
4. Detach and position aside the central junction box (CJB) by removing one screw on the left side and disengaging the retainer clip on the right side.
5. Using a small pick tool, remove the clips and disconnect the fluid tubes from the master cylinder. See Figure 1.

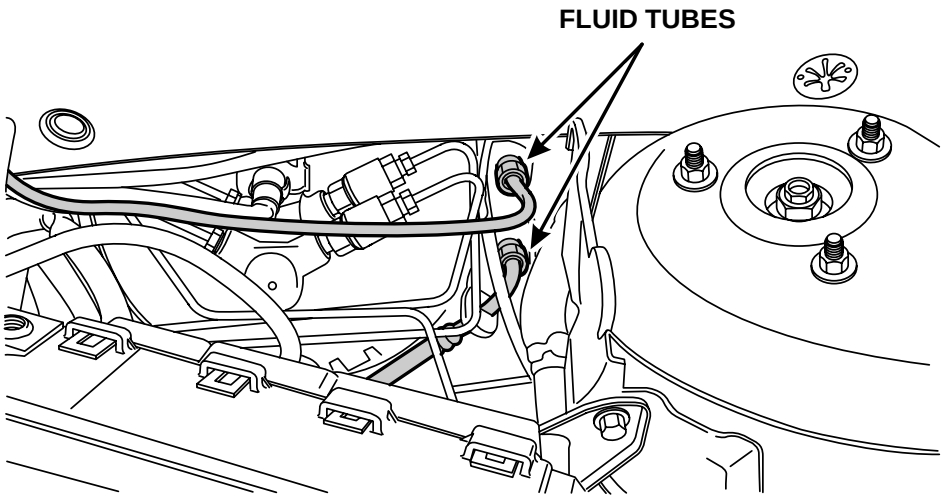



**FIGURE 1**

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6. Remove the instrument panel lower finish panel retainer screws. Then loosen the hood release cable handle nut and position aside the cable and data link connector. Remove the finish panel. See Figure 2.

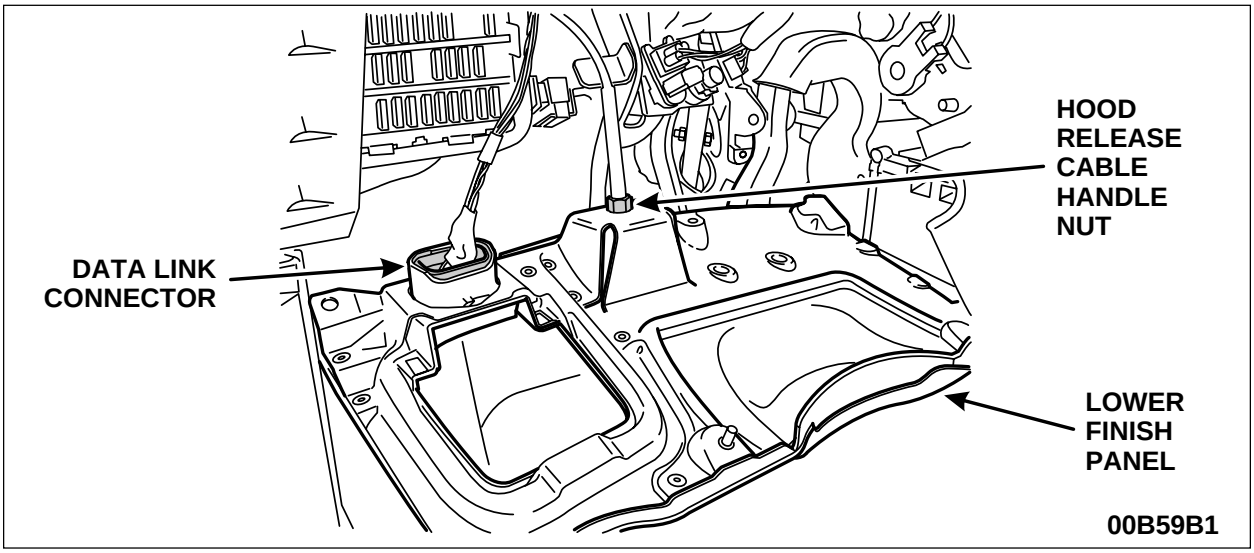



**FIGURE 2**

7. Remove the clip securing the fluid tube bracket at the dash panel. See Figure 3.
8. Remove the brake pedal pushrod retainer pin.
9. At the pedal assembly, disconnect the electrical connectors from the switches, and then remove the switches from the assembly. See Figure 3.

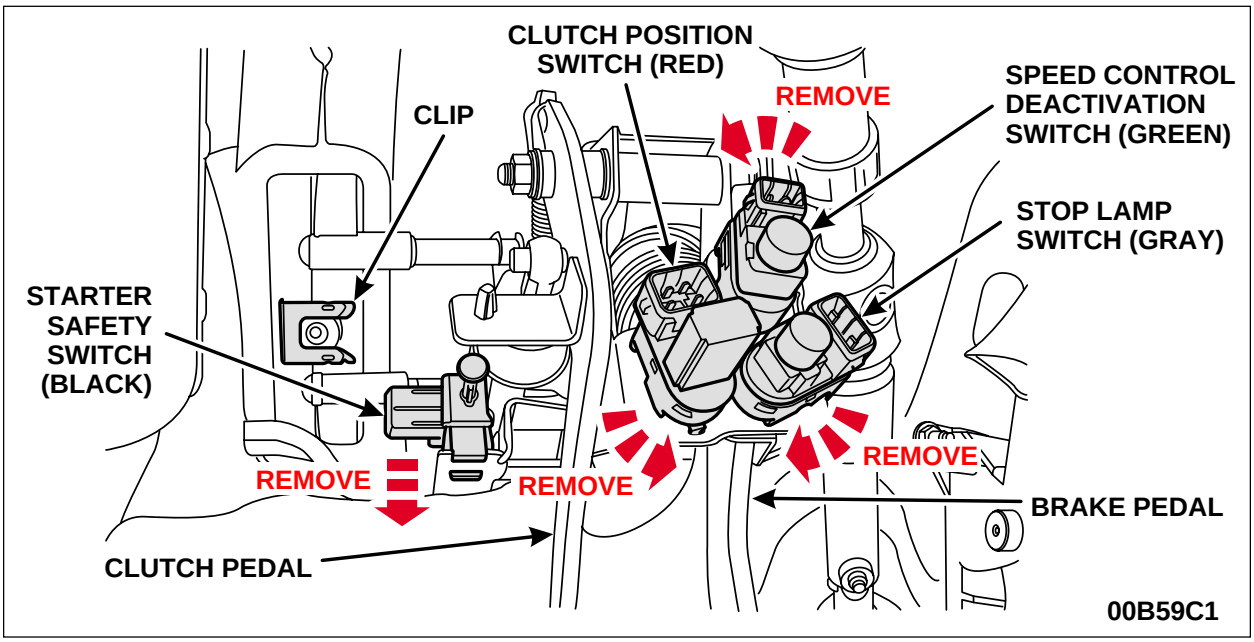



**FIGURE 3**

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10. **CAUTION: The complete clutch/brake pedal assembly must be removed from the vehicle to gain adequate clearance to properly tighten the bolts when installing the new clutch master cylinder.**

    Remove the pedal assembly from the vehicle and mount it in a vise. See Figure 4.

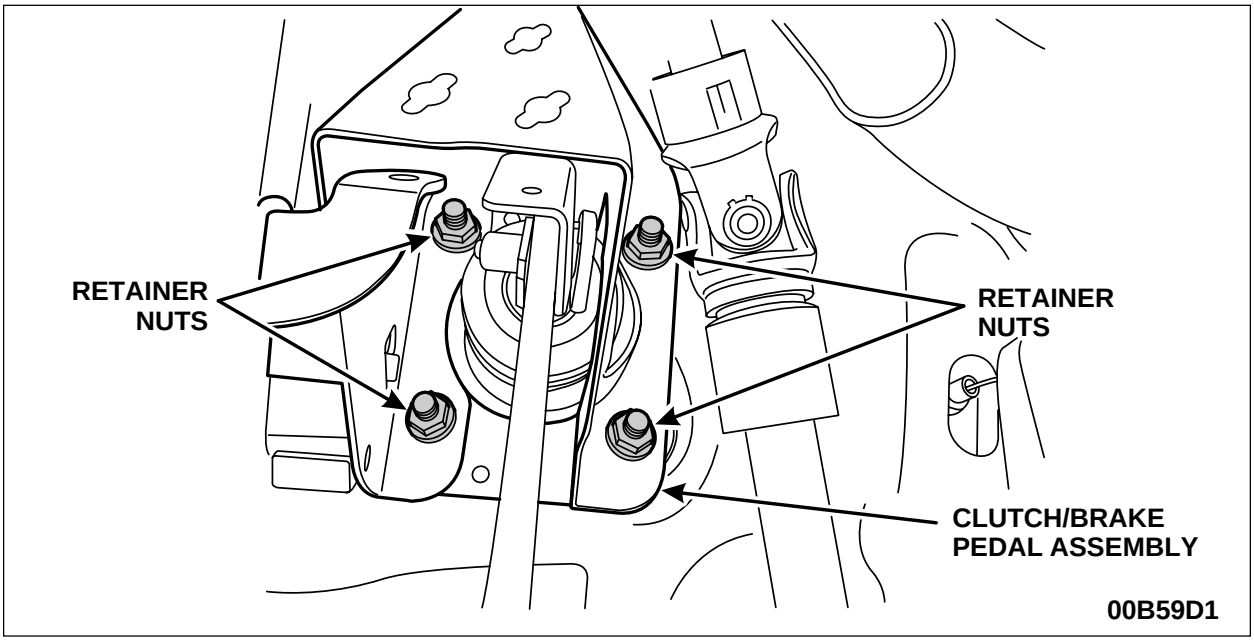



**FIGURE 4**

11. Remove the clutch master cylinder retainer bolts, then detach the master cylinder from the pedal assembly and slide the rod off the pedal arm. See Figure 5.

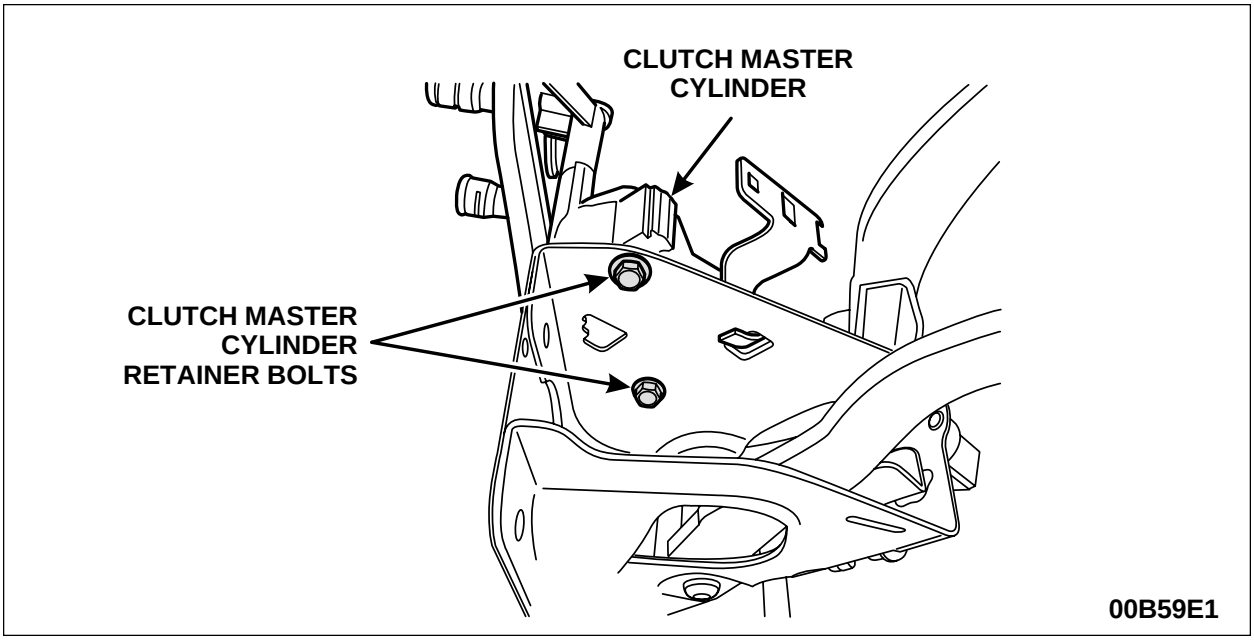



**FIGURE 5**

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12. Remove the clutch pedal return spring. See Figure 6.

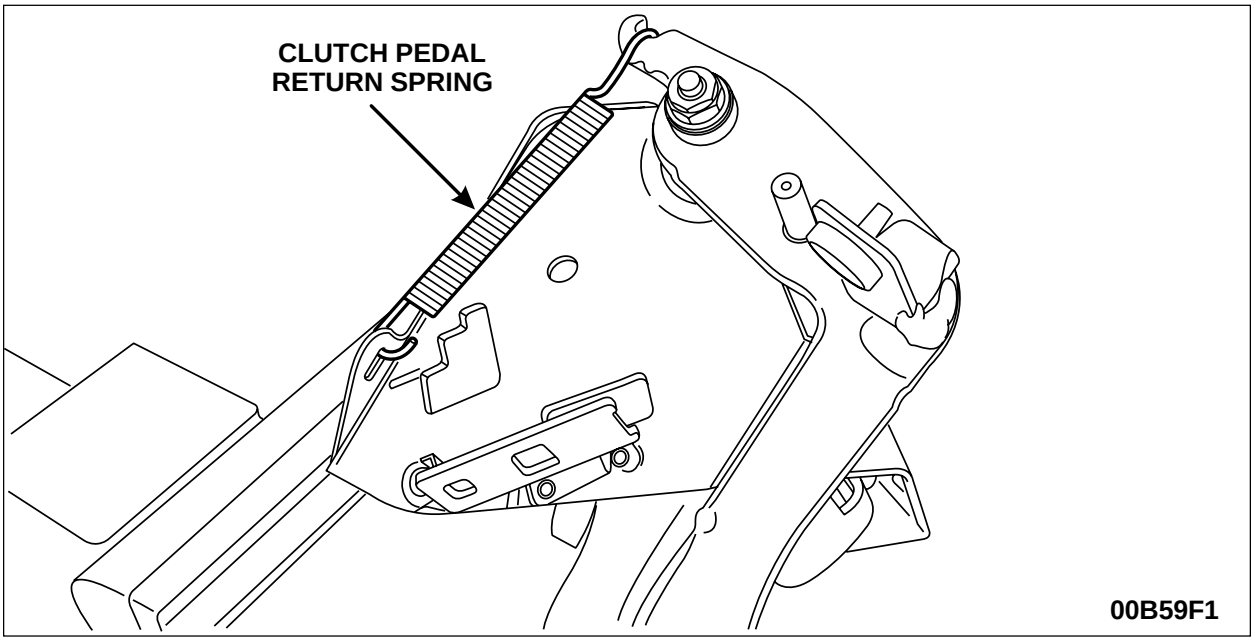



**FIGURE 6**

13. Destroy, then discard the removed master cylinder and spring.

## INSTALLATION

1. Install the new clutch pedal return spring.
2. Position the new clutch master cylinder to the pedal support and attach the rod to the pedal arm. Attach the master cylinder onto the pedal support bracket and install the retainer bolts. Tighten the bolts to 9 Nm (80 lb-in).

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3. NOTE: Once the brake pedal pushrod retaining pin is installed, make sure it is locked in position by trying to push the retaining pin back out of the pedal/pushrod assembly without depressing the locking tab. When properly installed, the locking tab on the side of the retaining pin will be visible as it extends through the pushrod and the brake pedal. See Figure 7.

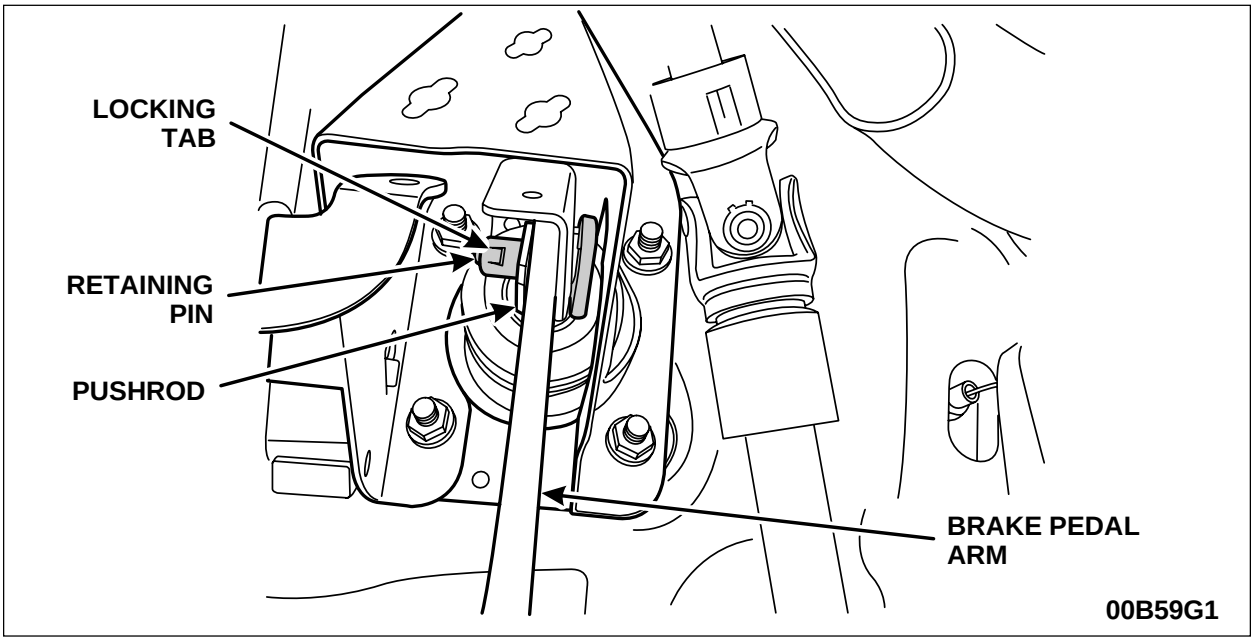



**FIGURE 7**

Position the pedal assembly in the vehicle. Attach the brake pedal pushrod to the brake pedal arm by installing the retainer pin. Install the pedal assembly retainer nuts and tighten them to 23 Nm (17 lb-ft).

4. Install the clip to secure the fluid tube bracket at the dash panel.

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5. Except for the red clutch pedal position switch, pull the plunger of each pedal assembly switch out to its fullest extended position, preparing the switches for adjustment. To make sure the plunger is at its fullest extended position, measure the plunger from point A to point B, as indicated in Figure 8. Then, compare that measurement to the plunger lengths in the Pedal Assembly Switches chart. Do this on all switches except for the clutch pedal position switch (red in color, part number 7C534). This switch is NOT adjustable. See Figure 8.

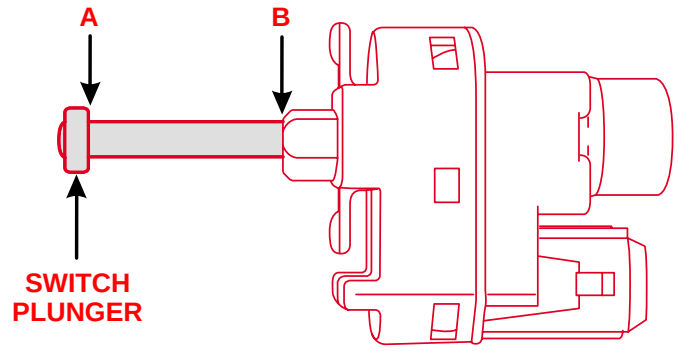



**FIGURE 8**

**PEDAL ASSEMBLY SWITCHES**

| Color | Part Number | Application | Plunger Length (fully extended) |
|---|---|---|---|
| Gray | 13480 | Stop Lamp | 21 mm (7/8 in.) |
| Green | 9C872 | Speed control deactivation | 24 mm (1 in.) |
| Black | 11A152 | Starter safety | 26 mm (1 1/16 in.) |
| Red | 7C534 | Clutch pedal position (PCM input) | Not adjustable |

6. **CAUTION: When installing the electrical switches, the stop lamp switch is rotated to the left (counterclockwise) and the remaining switches are rotated to the right (clockwise). Failure to do this will cause the plunger to bind inside the switch.**

   NOTE: The speed control deactivator switch (green) and the stop lamp switch (gray) will be adjusted as they are installed. A slight “ratcheting” noise and/or feel during switch adjustment is normal.

   Install all switches into the pedal assembly and connect all electrical connectors. Check the switches to make sure that they are secure and will not fall out.

7. NOTE: You need only to apply normal pressure to the clutch pedal for switch adjustment. A slight “ratcheting” noise and/or feel during switch adjustment is normal.

   Adjust the starter safety switch (black) by pressing and then releasing the clutch pedal.

8. Inspect the fluid tube O-rings and replace if damaged.

9. Remove the shipping plugs and make sure both retainer clips are in place, then connect the fluid tubes to the clutch master cylinder.

10. Install the central junction box.

11. Install the air cleaner assembly. Connect the MAFS electrical connector and the crankcase hose. Secure the air cleaner outlet tube to the throttle body with a suitable clamp.

12. Proceed to the appropriate bleed procedure.

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## BLEED PROCEDURE – MTX75

***WARNING!***

***BRAKE FLUID CONTAINS POLYGLYCOL ETHERS AND POLYGLYCOLS. AVOID CONTACT WITH THE EYES. WASH HANDS THOROUGHLY AFTER HANDLING. IF BRAKE FLUID CONTACTS THE EYES, FLUSH FOR 15 MINUTES WITH COLD RUNNING WATER. GET MEDICAL ATTENTION IF IRRITATION PERSISTS. IF TAKEN INTERNALLY, DRINK WATER AND INDUCE VOMITING. GET MEDICAL ATTENTION IMMEDIATELY. FAILURE TO FOLLOW THESE INSTRUCTIONS MAY RESULT IN PERSONAL INJURY.***

**CAUTION: If brake fluid is spilled on any painted surface, the affected area must be immediately washed with cold water.**

NOTE: Do not reuse any brake fluid.

NOTE: Continually check the brake fluid level while bleeding the system and top it off as necessary. Use only Super DOT 4 Brake Fluid or equivalent meeting Ford specification ESD-M6C57-A.

1. Fill the brake fluid reservoir to the top edge with the specified brake fluid and install the cap.
2. Remove the rubber cap from the bleeder screw at the slave cylinder, then loosen and retighten the screw. See Figure 9.

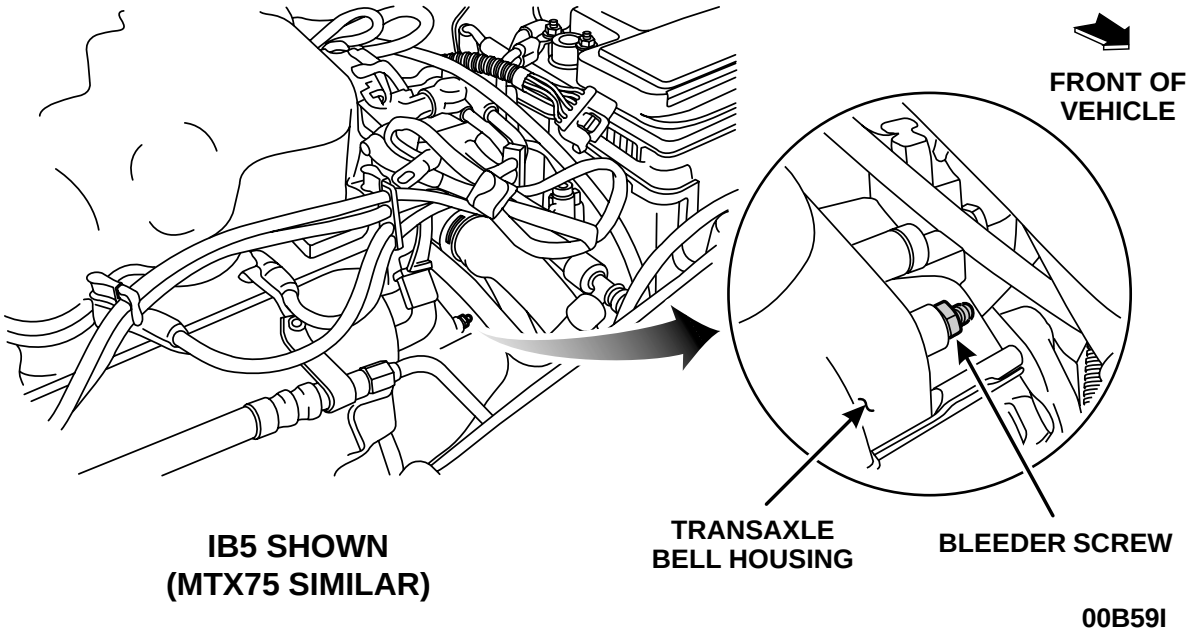



**FIGURE 9**

3. Attach a clear hose to the bleeder screw with the other end of the hose set in a drain pan or other suitable container.

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4. With the help of an assistant, bleed the clutch master cylinder as follows:
   a) Slowly depress the clutch pedal several times and then hold it to the floor.
   b) Open the bleeder screw and allow the air to escape, then close the screw.
   c) Repeat until no more air bubbles escape from the master cylinder, topping off the fluid level in the reservoir as necessary.
   d) Tighten the bleeder screw to 14 Nm (10 lb-ft) and install the rubber cap.
5. Recheck the brake fluid level, making sure the level is at the MAX mark. Install the cap.
6. Check clutch operation.

## BLEED PROCEDURE – IB5

**Tools Required: Vacuum Pump Kit 416-002D or equivalent with holding reservoir and pressure output capability**

***WARNING!***

***BRAKE FLUID CONTAINS POLYGLYCOL ETHERS AND POLYGLYCOLS. AVOID CONTACT WITH THE EYES. WASH HANDS THOROUGHLY AFTER HANDLING. IF BRAKE FLUID CONTACTS THE EYES, FLUSH FOR 15 MINUTES WITH COLD RUNNING WATER. GET MEDICAL ATTENTION IF IRRITATION PERSISTS. IF TAKEN INTERNALLY, DRINK WATER AND INDUCE VOMITING. GET MEDICAL ATTENTION IMMEDIATELY. FAILURE TO FOLLOW THESE INSTRUCTIONS MAY RESULT IN PERSONAL INJURY.***

**CAUTION: If brake fluid is spilled on any painted surface, the affected area must be immediately washed with cold water.**

NOTE: Do not reuse any brake fluid.

NOTE: Continually check the brake fluid level while bleeding the system and top it off as necessary. Use only Super DOT 4 brake fluid or equivalent meeting Ford specification ESD-M6C57-A.

1. Check to make sure the level in the brake fluid reservoir is at or near the MIN mark.
2. Remove the rubber cap from the bleeder screw at the slave cylinder, then loosen and retighten the screw. See Figure 7.
3. Fill the reservoir of the Vacuum Pump Kit with approximately 100 ml (3-4 ounces) Super DOT 4 brake fluid.
4. NOTE: The reservoir of the vacuum pump kit must be kept below the level of the bleeder screw.

   Attach the hose from the reservoir to the bleeder screw. Loosen the bleeder screw and pump approximately 80 ml (2.7 ounces) of brake fluid into the clutch control system. Then, tighten the bleeder screw.
5. Fill the brake fluid reservoir to the MAX mark with Super DOT 4 brake fluid and install the cap.
6. Tighten the bleeder screw to 14 Nm (10 lb-ft) and install the rubber cap.
7. Cycle the clutch pedal slowly 4-5 times.
8. Check clutch pedal operation.

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